

GRAND FINALE • A DATASET CALLED “EXCLUDED”

A new 1,197-image set, investigated before anything was built on it

Re-ran our clustering pipeline — K=6 (of Phase 2) was too coarse, so we iterated to K=10 and re-split three still-mixed clusters, arriving at 13 categories confirmed by eye.

1,197

images, spec-verified before any training began

13

verified categories, up from Phase 2's 7

2 new categories

biological / cellular SEM content — never present in any prior round

Same discipline as before: 10% held out per category, so the two brand-new cellular categories are represented in validation, not left to chance.

GRAND FINALE • TESTED, NOT ASSUMED

Don't start over — adapt what already works

Three strategies, measured head-to-head on the same held-out set, across all 13 categories:

From scratch, new data only

1,083 images, trained fresh

23.439 dB

Untouched Phase 2 model

Never trained on this data — beat the from-scratch model on all 13 categories

23.664 dB

Fine-tuned from Phase 2 checkpoint

Gentle, low learning rate — beat the old model on 12 of 13 categories

23.697 dB

1,083 new images alone can't match a model that already learned from 4,310 — fine-tuning wins.

Rejecting augmentation, and ruling out catastrophic forgetting

Augmentation, tried twice, rejected

Synthetic-noise augmentation helped in Phase 2 — here it made fine-tuning worse at two learning rates (23.631, 23.649 vs. 23.697). Rotation alone, no synthetic data, came close (23.682): doubling the pool with synthetic images diluted focused exposure to real images during a deliberately gentle fine-tune.

No catastrophic forgetting

Domain	Old	Fine-tuned
Phase 1 (natural photos)	27.610	27.736
Phase 2 (original SEM domain)	23.664	23.611

A tiny, noise-level dip on old data — the model gained new capability without losing what it already had.

GRAND FINALE · THROUGHPUT

A 1.95× speedup, verified to cost zero quality

Baseline · run.py, batch-8, T4, FP32

4.721 ms



FP16, patched into run.py itself

2.419 ms

Speedup

1.95×

How

Casts model + batch to FP16 automatically when a GPU is present; CPU-only environments fall back to FP32, since FP16 isn't reliably faster there.

Verified, not assumed

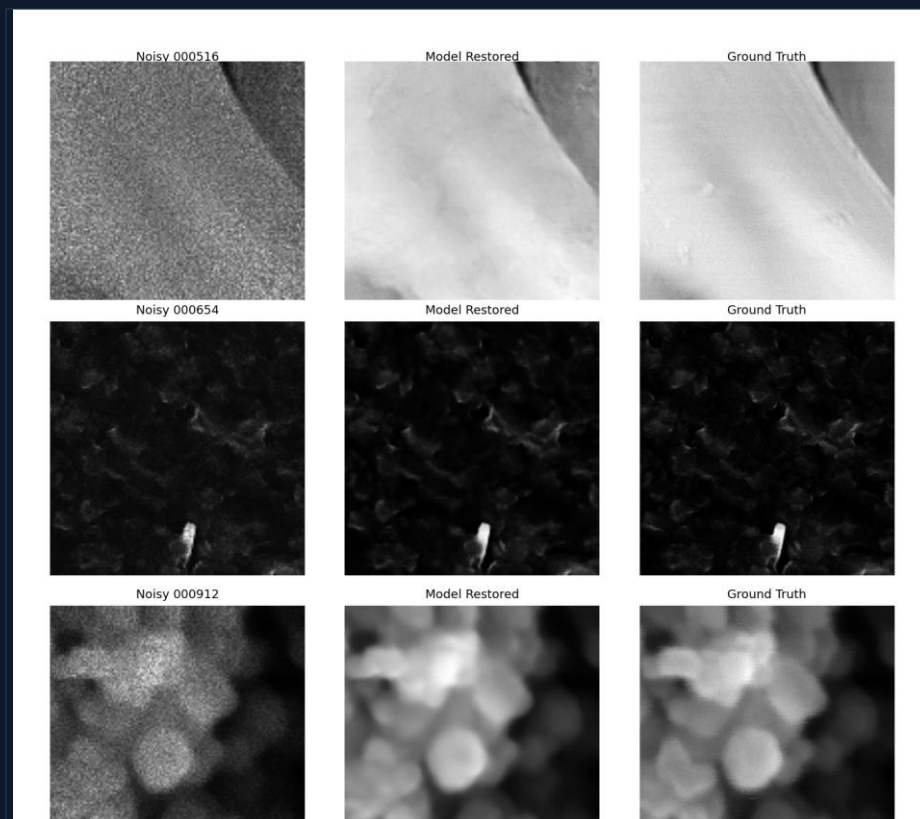
Full 1,197-image run: PSNR/SSIM identical to FP32 to 3–4 decimals (23.697 dB / 0.6238). 0 contract violations across all outputs.

INT8, ruled out

Measured honestly: ran on CPU, ~62× slower than FP16-GPU. Correctly not pursued.

GRAND FINALE · BEST RESTORATIONS

Highest-PSNR cases, on the new 13-category data



The fine-tuned checkpoint on unseen, more diverse content — including morphologies well outside Phase 2's original 7 categories.